Chapter 3: Vector Analysis

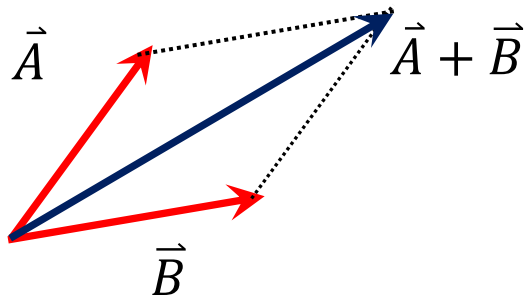
The following subjects will be discussed

- Vector Representation.
- Basic Vector Operations.
- Examples.

Basic Vector Operations

Addition/subtraction

$$\vec{A} \pm \vec{B} = (A_x \pm B_x)\,\vec{\imath} + (A_y \pm B_y)\,\vec{\jmath} + (A_z \pm B_z)\,\vec{k}$$



$$\vec{A} + \vec{B} = \vec{B} + \vec{A} \qquad \vec{B} - \vec{A} = -(\vec{A} - \vec{B})$$

$$\vec{A} + \vec{B} + \vec{C} = (\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$$

$$k(\vec{A} \pm \vec{B}) = k\vec{A} \pm k\vec{B}$$

Example Given $\vec{A} = 10\,\vec{\imath} - 4\,\vec{\jmath} + 6\,\vec{k}$ $\vec{B} = 2\,\vec{\imath} + \vec{\jmath}$

1) The component of $\vec{A}$ along the y-direction -4

2) The magnitude of $3\vec{A} - \vec{B}$

$$3\vec{A} - \vec{B} = 30\,\vec{\imath} - 12\,\vec{\jmath} + 18\,\vec{k} - 2\,\vec{\imath} - \vec{\jmath} = 28\,\vec{\imath} - 13\,\vec{\jmath} + 18\,\vec{k}$$

$$\left|3\vec{A} - \vec{B}\right| = \sqrt{(28)^2 + (13)^2 + (18)^2} = \sqrt{1277}$$

Basic Vector Operations

Example

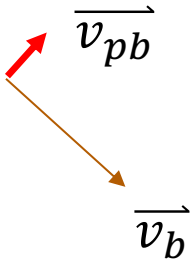
$$\overrightarrow{v_b} = 10(\cos(-45)\ \vec{i} + \sin(-45)\ \vec{j}\ )$$

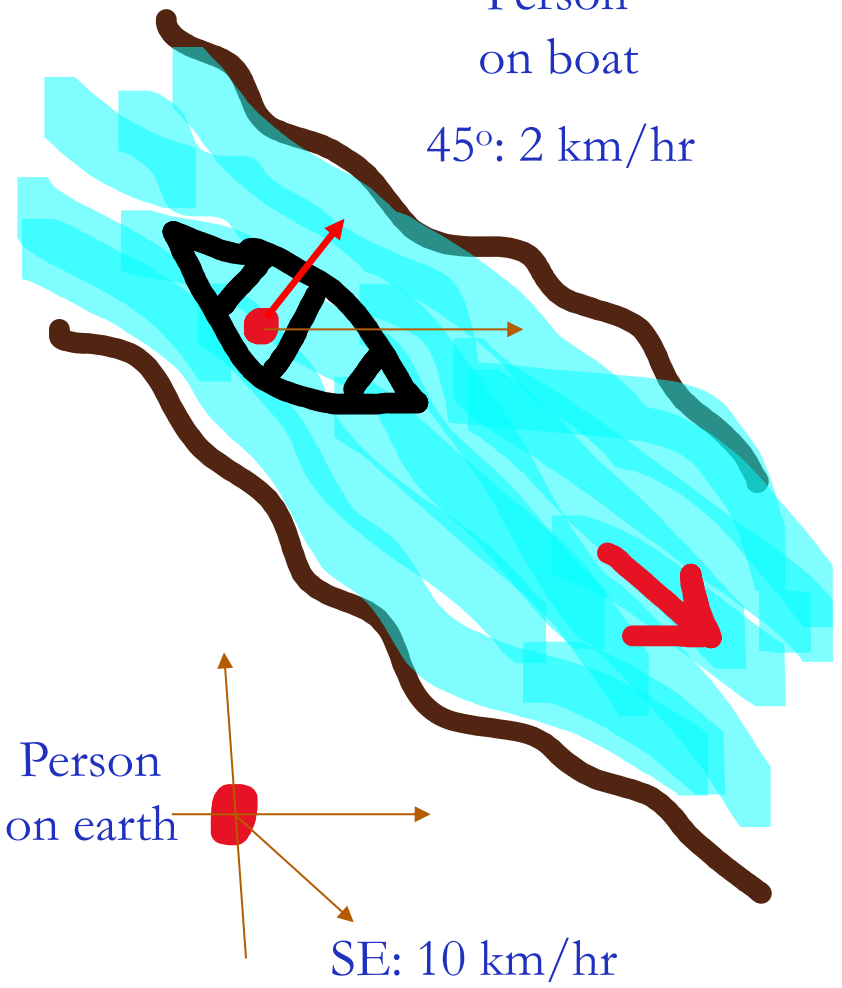
$$\overrightarrow{v_{pb}} = 2(\cos(45)\ \vec{i} + \sin(45)\ \vec{j}\ )$$

$$\overrightarrow{v_p} = \overrightarrow{v_b} + \overrightarrow{v_{pb}}$$

$$\overrightarrow{v_p} = 8.48\ \vec{i} - 5.66\vec{j} \quad \text{km/hr}$$

$$\overrightarrow{v_p} = 10.2 \angle -33.72^{o} \quad \text{km/hr}$$





Basic Vector Operations

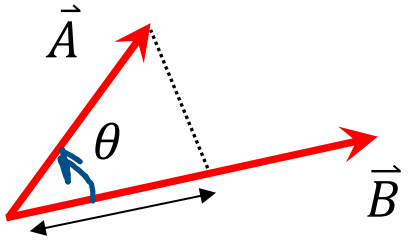
Scalar Multiplication (dot product)

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|\cos\theta$$

$$\vec{A} \cdot \vec{B} = (A_x \cdot B_x) + (A_y \cdot B_y) + (A_z \cdot B_z)$$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

Scalar result



Shadow (projection) of $\vec{A}$ on $\vec{B}$

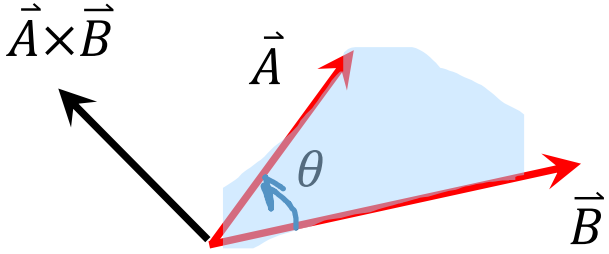
$|\vec{A}|\cos\theta$

Vector Multiplication (cross product)

$$\vec{A}\times\vec{B} = |\vec{A}||\vec{B}|\sin\theta\ \overrightarrow{a_n}$$

Vector result

$$\vec{A}\times\vec{B} = \begin{vmatrix} \vec{\imath} & \vec{\jmath} & \vec{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$



$$\vec{A}\times\vec{B} = (A_yB_z - A_zB_y)\vec{\imath} - (A_xB_z - A_zB_x)\vec{\jmath} + (A_xB_y - A_yB_x)\vec{k}$$

$$\vec{A}\times\vec{B} \neq \vec{B}\times\vec{A}$$

Basic Vector Operations

Example

Given $\vec{A} = 3\,\vec{\imath} + 4\,\vec{\jmath} + \vec{k}$ $\vec{B} = 2\,\vec{\jmath} - 5\,\vec{k}$

Find θ_{AB}.

$$\vec{A} \cdot \vec{B} = (3 \cdot 0) + (4 \cdot 2) + (1 \cdot -5)$$

$$\vec{A} \cdot \vec{B} = 3$$

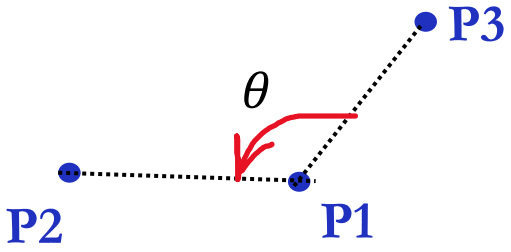
$$|\vec{A}| = \sqrt{26} \qquad |\vec{B}| = \sqrt{29}$$

$$\cos\theta_{AB} = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|} \qquad \theta_{AB} = 83.7°$$

Basic Vector Operations

Example Show that the points **P1** (5, 2, -4), **P2** (1, 1, 2) and **P3** (-3, 0, 8) all lie on a straight line.

Let's check the angle θ. If $\theta = 180°$, the three points lie on a straight line



$$\overrightarrow{P_1P_2} = -4\,\vec{\imath} - \vec{\jmath} + 6\,\vec{k} \qquad \left|\overrightarrow{P_1P_2}\right| = \sqrt{53}$$

$$\overrightarrow{P_1P_3} = -8\,\vec{\imath} - 2\,\vec{\jmath} + 12\,\vec{k} \qquad \left|\overrightarrow{P_1P_3}\right| = \sqrt{212}$$

$$\overrightarrow{P_1P_2} \cdot \overrightarrow{P_1P_3} = 32 + 2 + 72 = 106$$

$$\vec{A} \cdot \vec{B} = \left|\vec{A}\right|\left|\vec{B}\right|\cos\theta \quad \Rightarrow \quad \cos\theta = \frac{\vec{A} \cdot \vec{B}}{\left|\vec{A}\right|\left|\vec{B}\right|} = 1 \quad \Rightarrow \quad \theta = 0, 180°$$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} = \begin{vmatrix} \vec{\imath} & \vec{\jmath} & \vec{k} \\ -4 & -1 & 6 \\ -8 & -2 & 12 \end{vmatrix} = 0$$

Chapter 4: Multi-dimensional Motion

(Constant Acceleration)

The following subjects will be discussed

- Equations of motion in 3D.
- Examples.
- Special cases.

Multi-dimensional Motion

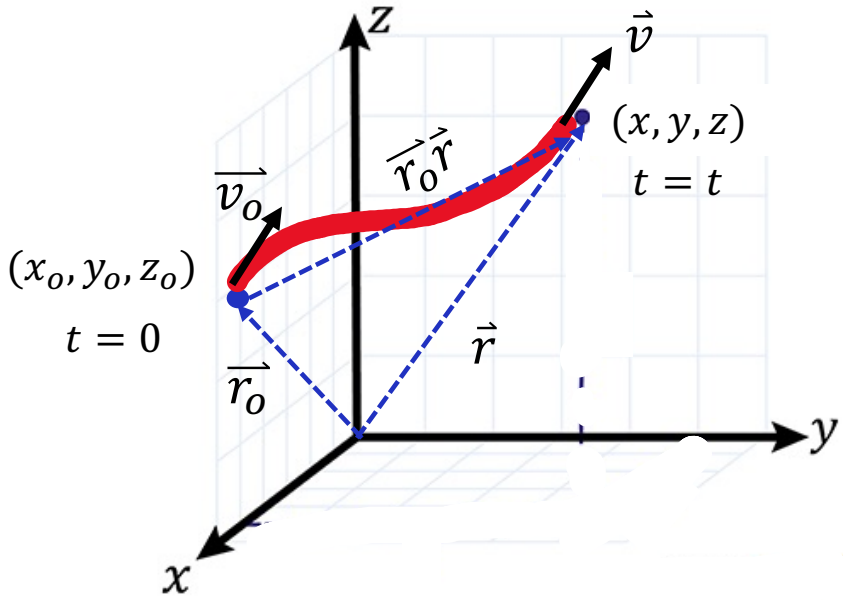
The approach is based on decomposing the motion into three independent 1D motions.

Initial position

$$\vec{r_o} = x_o\,\vec{i} + y_o\,\vec{j} + z_o\,\vec{k}$$

Initial velocity

$$\vec{v_o} = v_{ox}\,\vec{i} + v_{oy}\,\vec{j} + v_{oz}\,\vec{k}$$



At any time

$$\vec{r} = x(t)\,\vec{i} + y(t)\,\vec{j} + z(t)\,\vec{k}$$

Each can be studied independently

$$\vec{r_o}\vec{r} = (x - x_o)\,\vec{i} + (y - y_o)\vec{j} + (z - z_o)\,\vec{k}$$

$$\vec{v} = v_x(t)\,\vec{i} + v_y(t)\,\vec{j} + v_z(t)\,\vec{k}$$

$$\vec{a} = a_x(t)\,\vec{i} + a_y(t)\,\vec{j} + a_z(t)\,\vec{k}$$

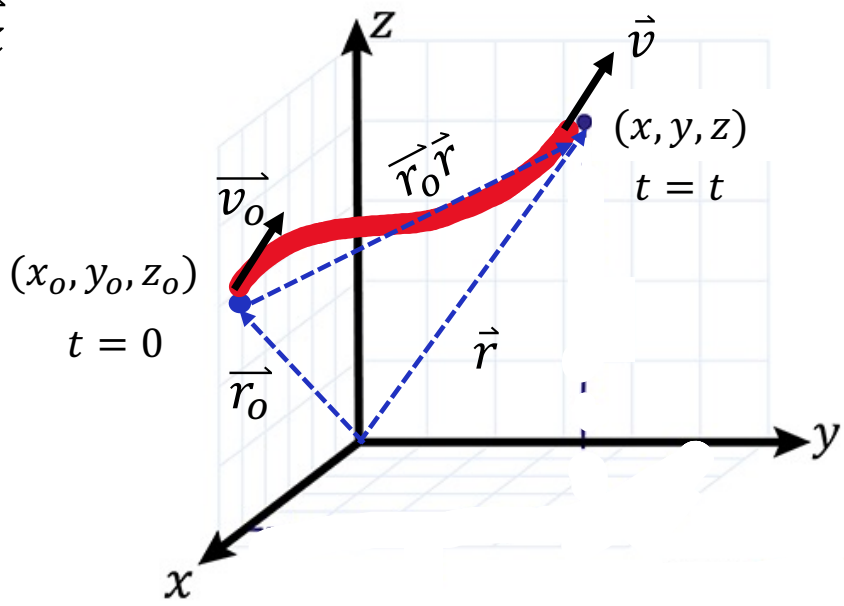
3D Equations of Motion

Average velocity

$$\vec{v}_{ave} = \frac{\Delta\vec{r}}{\Delta t} = \underbrace{\frac{x - x_o}{t - t_o}}_{v_{x_ave}}\vec{i} + \underbrace{\frac{y - y_o}{t - t_o}}_{v_{y_ave}}\vec{j} + \underbrace{\frac{z - z_o}{t - t_o}}_{v_{z_ave}}\vec{k}$$

Instantaneous velocity

$$\vec{v} = \frac{d\vec{r}}{dt} = \underbrace{\frac{dx}{dt}}_{v_x}\vec{i} + \underbrace{\frac{dy}{dt}}_{v_y}\vec{j} + \underbrace{\frac{dz}{dt}}_{v_z}\vec{k}$$



Average acceleration

$$\vec{a}_{ave} = \frac{\Delta\vec{v}}{\Delta t} = \underbrace{\frac{v_x - v_{ox}}{t - t_o}}_{a_{x_ave}}\vec{i} + \underbrace{\frac{v_y - v_{oy}}{t - t_o}}_{a_{y_ave}}\vec{j} + \underbrace{\frac{v_z - v_{oz}}{t - t_o}}_{a_{z_ave}}\vec{k}$$

Instantaneous acceleration

$$\vec{a} = \frac{d\vec{v}}{dt} = \underbrace{\frac{dv_x}{dt}}_{a_x}\vec{i} + \underbrace{\frac{dv_y}{dt}}_{a_y}\vec{j} + \underbrace{\frac{dv_z}{dt}}_{a_z}\vec{k}$$

3D Equations of Motion

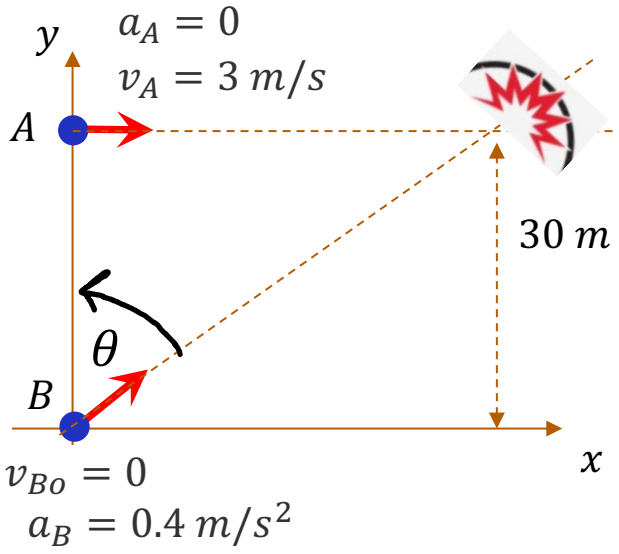
Motion in x direction	Motion in y direction	Motion in z direction
$v_x(t) = a_x t + v_{ox}$	$v_y(t) = a_y t + v_{oy}$	$v_z(t) = a_z t + v_{oz}$
$x(t) = x_o + v_{ox} t + \frac{1}{2} a_x t^2$	$y(t) = y_o + v_{oy} t + \frac{1}{2} a_y t^2$	$z(t) = z_o + v_{oz} t + \frac{1}{2} a_z t^2$
$v_x^2(t) = v_{ox}^2 + 2a_x \cdot [x(t) - x_o]$	$v_y^2(t) = v_{oy}^2 + 2a_y \cdot [y(t) - y_o]$	$v_z^2(t) = v_{oz}^2 + 2a_z \cdot [z(t) - z_o]$
$x(t) = x_o + \frac{1}{2} [v_x(t) + v_{ox}] \cdot t$	$y(t) = y_o + \frac{1}{2} [v_y(t) + v_{oy}] \cdot t$	$z(t) = z + \frac{1}{2} [v_z(t) + v_{oz}] \cdot t$
$x(t) = x_o + v_x(t) \cdot t - \frac{1}{2} a_x t^2$	$y(t) = y_o + v_y(t) \cdot t - \frac{1}{2} a_y t^2$	$z(t) = z_o + v_z(t) \cdot t - \frac{1}{2} a_z t^2$

Example

What is the collision angle?

At the collision:

1) Both balls at the same x same y
2) Both reach at the same time.



Initial situation $t = 0$.

Ball A

$$x(t) = x_o + v_{ox}t + \frac{1}{2}a_x t^2$$

$$x_A = v_{ox}t = 3t$$

Ball B

$$a_{Bx} = 0.4\sin\theta \qquad a_{By} = 0.4\cos\theta$$

$$x(t) = x_o + v_{ox}t + \frac{1}{2}a_x t^2$$

$$x_B = \frac{1}{2}a_{Bx}t^2 = \frac{1}{2}0.4\sin\theta\, t^2$$

$$x_A = x_B \quad \Rightarrow \quad t = \frac{15}{\sin\theta}$$

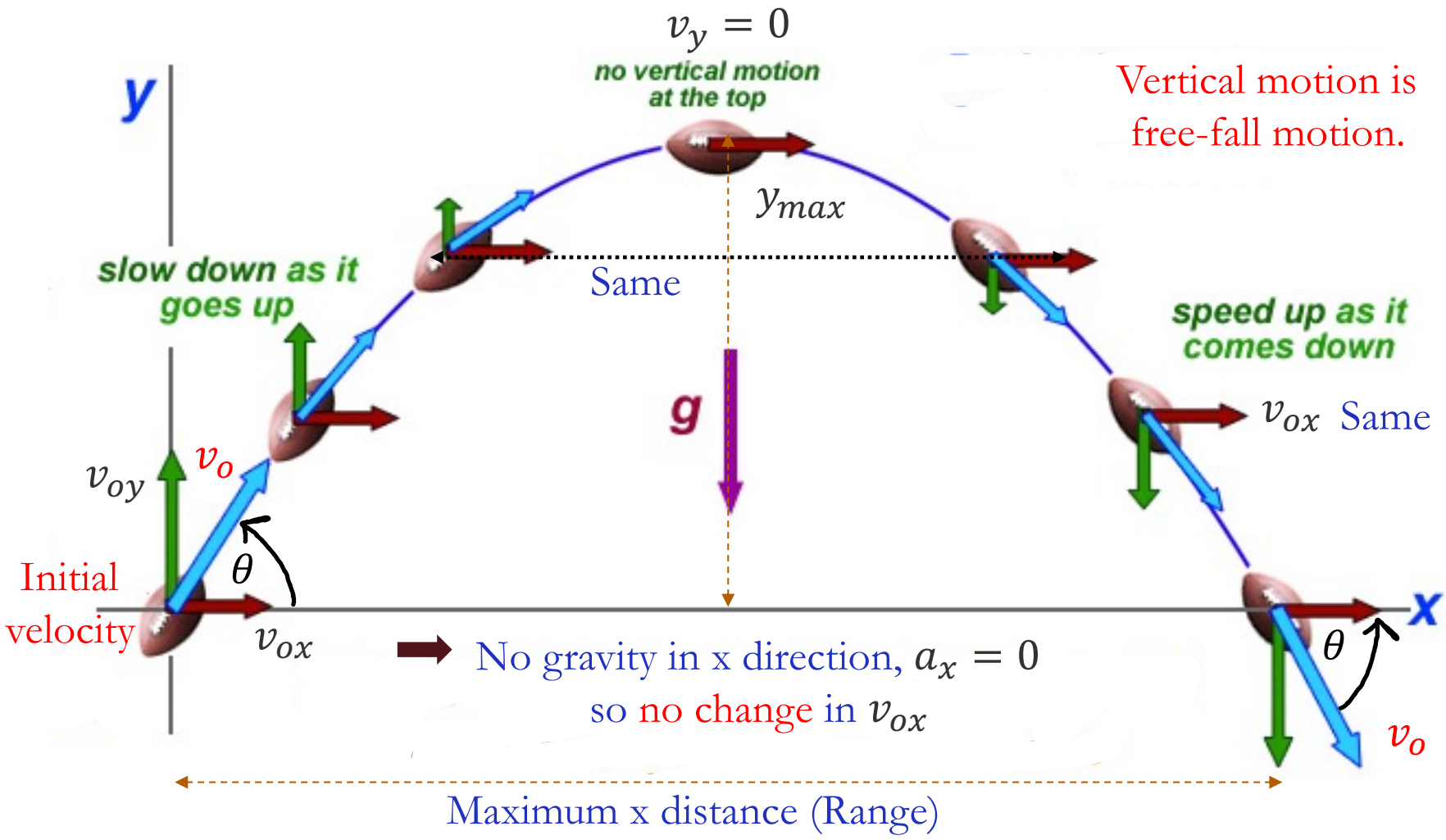
$$y_B = \frac{1}{2}a_{By}t^2 = \frac{1}{2}0.4\cos\theta\, t^2 = 30$$

$$\frac{1}{2}0.4\cos\theta\left(\frac{15}{\sin\theta}\right)^2 = 30$$

$$\theta = 60°$$

Projectile Motion

A 2D motion with no external forces (only gravity)



Projectile Motion

Analysis

$$v_{ox} = v_o \cos\theta$$
$$v_{oy} = v_o \sin\theta$$

At any time

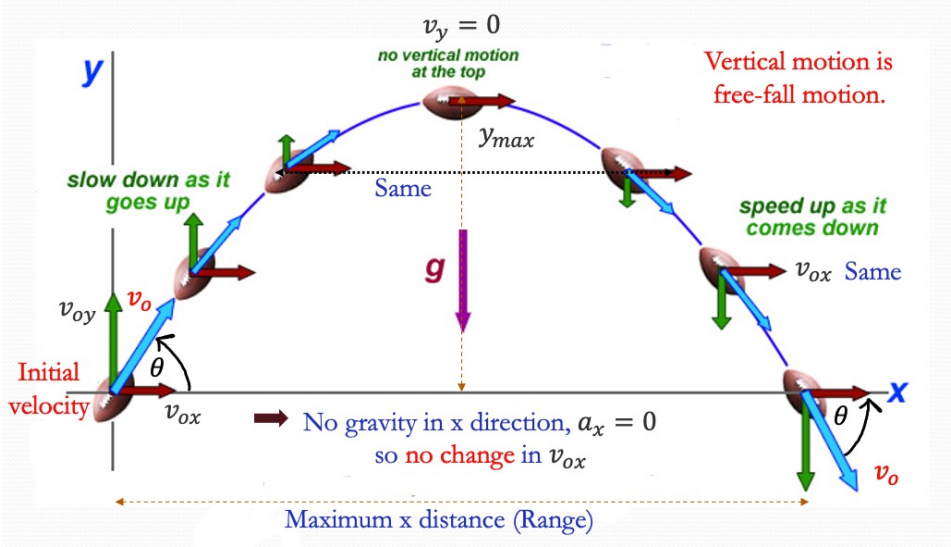
$$x(t) = x_o + v_{ox}t$$

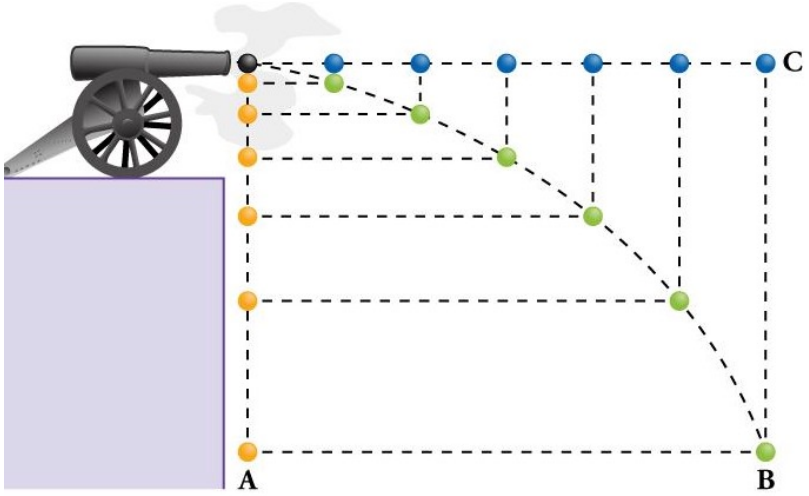
$$y(t) = y_o + v_{oy}t - \frac{1}{2}gt^2$$

$$x_{max} = Range = v_o \cdot \cos\theta \cdot t$$

$$R = \frac{{v_o}^2}{g} \cdot sin\,2\theta$$

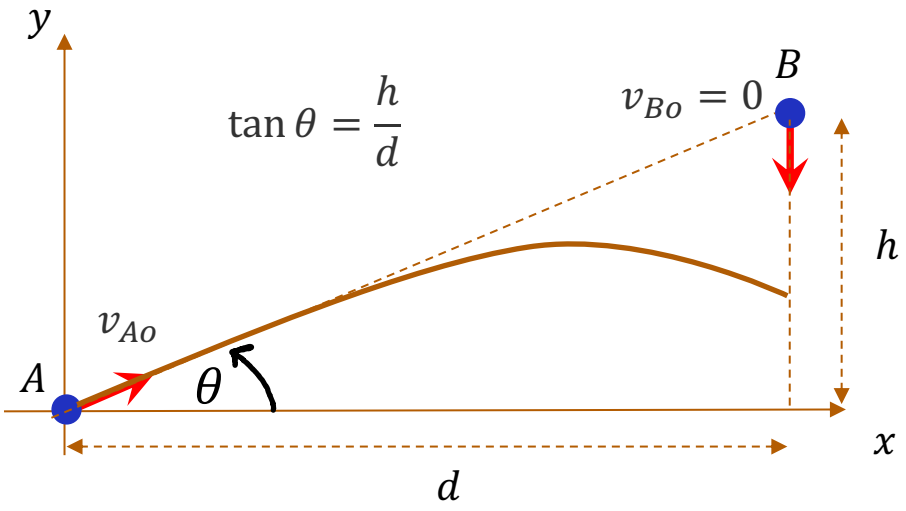
$$y_{max} = \frac{(v_o \sin\theta)^2}{2g}$$





Example

At what conditions will the balls collide mid-air?



Ball A: Projectile motion

$$x = x_o + v_{ox}t + \frac{1}{2}a_x t^2$$

$$d = v_{Aox}t$$

$$t = \frac{d}{v_{Aox}}$$

$$y = y_o + v_{oy}t - \frac{1}{2}gt^2$$

$$y_A = v_{Aoy}\frac{d}{v_{Aox}} - \frac{1}{2}gt^2$$

$$y_A = h - \frac{1}{2}gt^2$$

Ball B: Free-fall

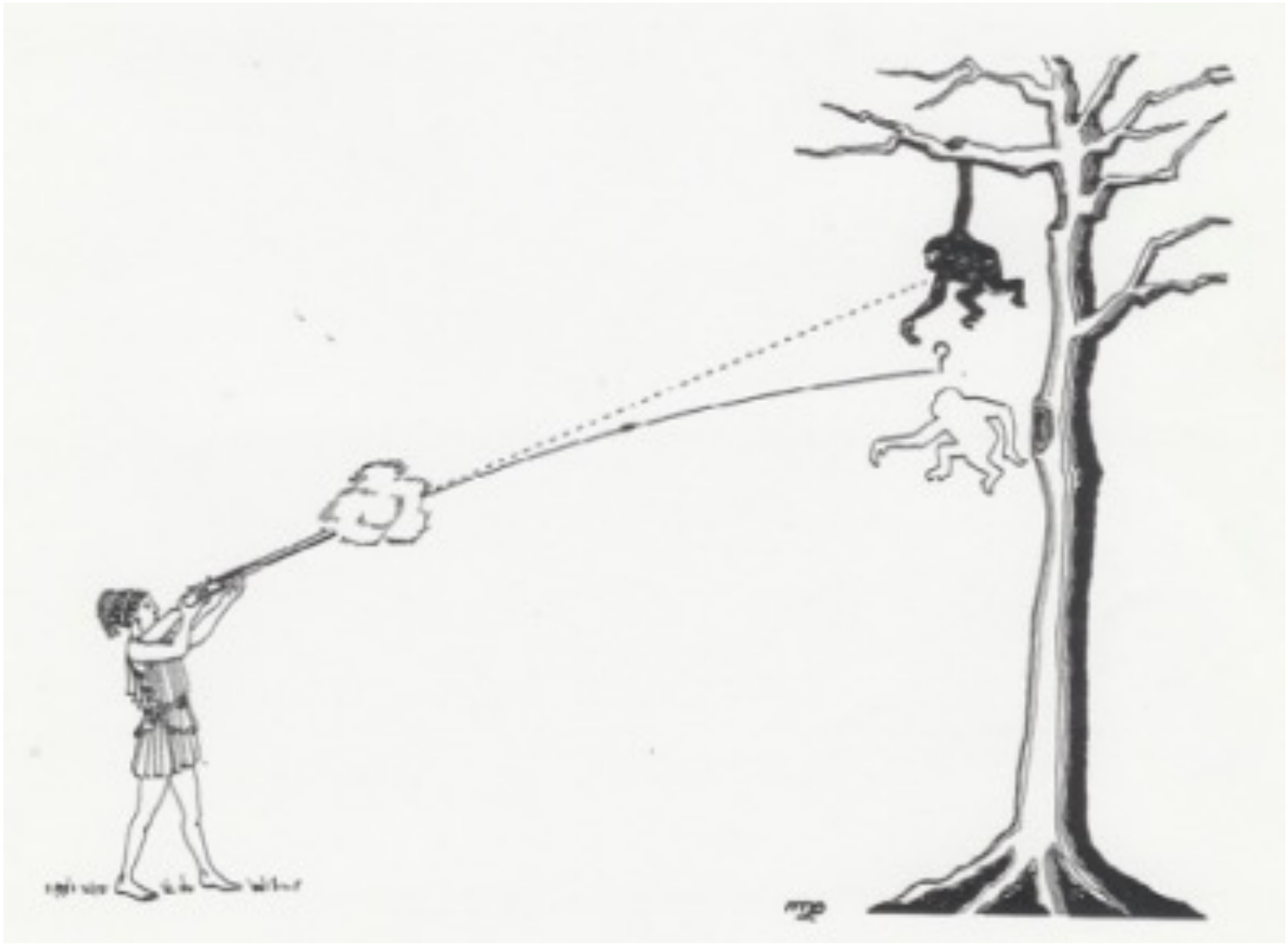
$$y = y_o + v_{oy}t - \frac{1}{2}gt^2$$

$$y_B = h - \frac{1}{2}gt^2$$

Conclusion

Balls will collide for any initial speed of A, as long as A is initially aimed at B.

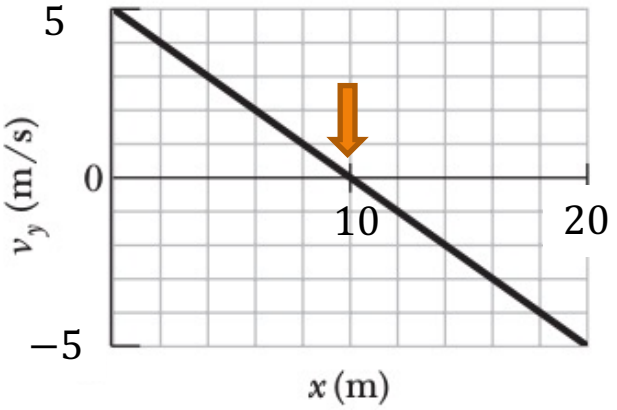
A (Sad) Example



Example

Find the angle to hit the bottom of the wall.





$$v_{oy} = 5\ m/s$$

$$v_o = \frac{v_{oy}}{\sin\theta} = \frac{5}{\sin\theta}$$

$$x = 10 = x_o + v_{ox}t + \frac{1}{2}a_x t^2$$

$$10 = v_o \cos\theta \cdot t$$

$$10 = \frac{5}{\sin\theta}\cos\theta \cdot t \qquad t = 2\tan\theta$$

$$v_y = 0 = v_{oy} - g\,t$$

$$5 = g\ 2\tan\theta$$

$$\tan\theta = \frac{5}{2g}$$

$$\theta = 14°$$

Example

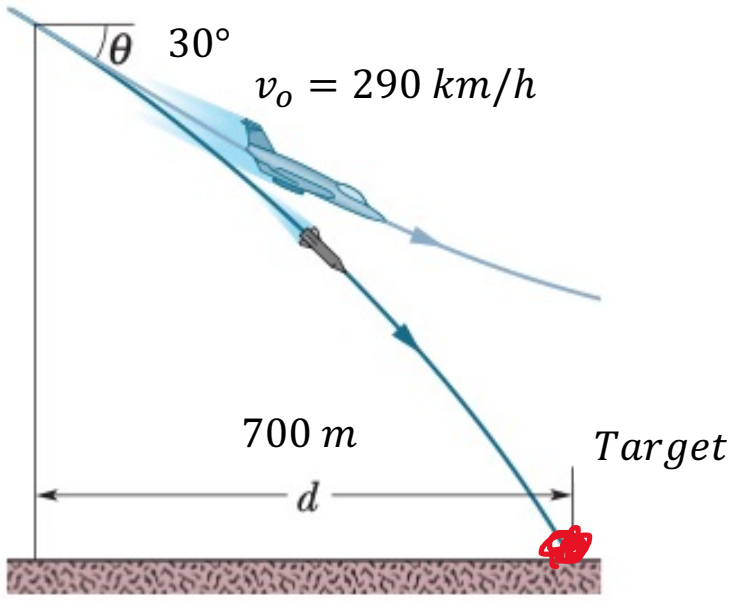
How long does the missile take to hit target?

$$v_{ox} = v_o \cos\theta = 290\cos(-30) = 69.8\ m/s$$

$$v_{oy} = v_o \sin\theta = 290\sin(-30) = -40.3\ m/s$$

$$x = 700 = x_o + v_{ox}t + \frac{1}{2}a_x t^2$$

$$700 = v_{ox}t \qquad t = \frac{700}{69.8} = 10\ s$$



How high was the release point?

$$y - y_o = v_{oy} \cdot t - \frac{1}{2} g\, t^2$$

$$-y_o = (-40.3)(10) - \frac{1}{2}(9.8)\,(10)^2$$

$$y_o = 893\ m$$

Q & A

Send me an email if
you have any
questions.